THE LOWER SHRIEK ALONG THE THIRD PROJECTION FOR A POLYNOMIAL ALGEBRA ON TOP

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement refers to the three-maps equation of the chapter, which is displayed earlier in the same chapter; the proof uses only the notation it establishes.

Editorial note. The seven references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), sliced mechanically from the chapter preamble.

Allowed standard prerequisites (editorial).

- Computation of the lower shriek via a cosimplicial resolution. Cohomology on Sites, Lemma sites-cohomology-lemma-compute-by-cosimplicial-resolution, used at proof lines 1453-1461. Reduces the claim to a homotopy statement about the mapping simplicial sets.
- Trivial Kan fibrations: base change, homotopy equivalence and filtered colimits. Simplicial, Lemmas simplicial-lemma-trivial-kan-base-change, simplicial-lemma-trivial-kan-homotopy and simplicial-lemma-filtered-colimit-trivial-kan, used at proof lines 1482-1503. Supply the two generating cases of the homotopy computation.
- Products of simplicial sets homotopy equivalent to a point. Simplicial, Lemma simplicial-lemma-products-homotopy, used at proof lines 1480-1481. Reduces the general case to the generating cases.
- The value of the lower shriek on the polynomial algebra. Lemma lemma-pi-lower-shriek-polynomial-algebra, used at proof lines 1459-1461. Identifies pi_{3,!}F with the constant simplicial group F(C -> C).

Source reference. Original equation 08QT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11071–11071. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08QG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11059–11059. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08Q5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11048–11048. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08Q4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11047–11047. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08NN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10996–10996. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08NS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11000–11000. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08Q9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11052–11052. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *With notation as in (08QT) suppose that C is a polynomial algebra over B. Then $L\pi_!(g_3^{-1}\mathcal{F}) = L\pi_{3,!}\mathcal{F} = \pi_{3,!}\mathcal{F}$ for any abelian sheaf $\mathcal{F}$ on $\mathcal{C}_{C/B}$*

Proof. Write $C = B[E]$ for some set E. Choose a resolution $P_\bullet \to B$ of B over A. For every n consider the object $U_n = (P_n \to B, P_n[E] \to C)$ of $\mathcal{C}_{C/B/A}$. Then $U_\bullet$ is a cosimplicial object of $\mathcal{C}_{C/B/A}$. Note that $u_3(U_\bullet)$ is the constant cosimplicial object of $\mathcal{C}_{C/B}$ with value $(C \to C)$. We will prove that the object $U_\bullet$ of $\mathcal{C}_{C/B/A}$ satisfies the hypotheses of Cohomology on Sites, Lemma 08Q9. This implies the lemma as it shows that $L\pi_!(g_3^{-1}\mathcal{F})$ is computed by the constant simplicial abelian group $\mathcal{F}(C \to C)$ which is the value of $L\pi_{3,!}\mathcal{F} = \pi_{3,!}\mathcal{F}$ by Lemma 08QG.

Let $U = (\beta : P \to B, \gamma : Q \to C)$ be an object of $\mathcal{C}_{C/B/A}$. We may write $P = A[S]$ and $Q = A[S \amalg T]$ by the definition of our category $\mathcal{C}_{C/B/A}$. We have to show that

$$\mathrm{Mor}_{\mathcal{C}_{C/B/A}}(U_\bullet, U)$$

is homotopy equivalent to a singleton simplicial set $*$. Observe that this simplicial set is the product

$$\prod\nolimits_{s\in S} F_s \times \prod\nolimits_{t\in T} F'_t$$

where F_s is the corresponding simplicial set for $U_s = (A[\{s\}] \to B, A[\{s\}] \to C)$ and F'_t is the corresponding simplicial set for $U_t = (A \to B, A[\{t\}] \to C)$. Namely, the object U is the product $\prod U_s \times \prod U_t$ in $\mathcal{C}_{C/B/A}$. It suffices each F_s and F'_t is homotopy equivalent to $*$, see Simplicial, Lemma 08Q4. The case of F_s follows as $P_\bullet \to B$ is a trivial Kan fibration (as a resolution) and F_s is the fibre of this map over $\beta(s)$. (Use Simplicial, Lemmas 08NN and 08NS). The case of F'_t is more interesting. Here we are saying that the fibre of

$$P_\bullet[E] \longrightarrow C = B[E]$$

over $\gamma(t) \in C$ is homotopy equivalent to a point. In fact we will show this map is a trivial Kan fibration. Namely, $P_\bullet \to B$ is a trivial can fibration. For any ring R we have

$$R[E] = \mathrm{colim}_{\Sigma\subset \mathrm{Map}(E,\mathbf{Z}_{\geq 0}) \text{ finite}} \prod\nolimits_{I\in\Sigma} R$$

(filtered colimit). Thus the displayed map of simplicial sets is a filtered colimit of trivial Kan fibrations, whence a trivial Kan fibration by Simplicial, Lemma 08Q5. □